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Assignment-5

Q.1 Objective: Configure Static Route in Cisco packet tracer

• Static route tells the device exactly where to send traffic, no matter what.

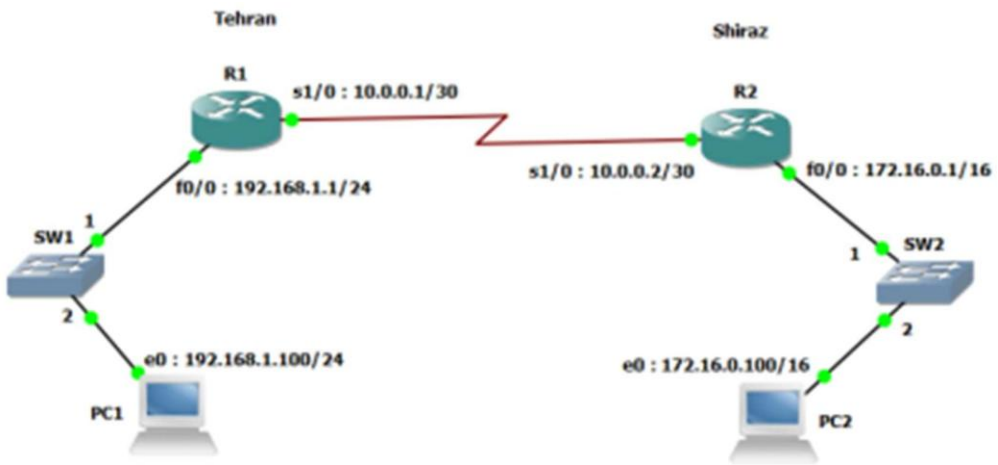
• Static route is often used when your network **has only a few routers** or **there is only one route from a source to a destination**.

Scenario

• Suppose that your company has **2 branches** located in **Tehran** and **Shiraz**. • As the administrator of the network, you are tasked to connect them so that employees in the two LANs **can communicate with each other**.

• After careful consideration you decided to connect them via **static**

route. **Physical Topology**



Physical Topology (for Packet Tracer)

Devices required:

- 2 × Routers (e.g., 2811)
- 2 × Switches (2960)
- 2 × PCs
- Serial DCE cable between routers
- Copper straight-through cables for LAN

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Commands:

```
R1> enable
R1# configure terminal
R1(config)# interface fastEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface serial 1/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
```

1. Configure IP address on Router R2 interfaces

Commands:

```
R2> enable
R2# configure terminal
R2(config)# interface serial 1/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# interface fastEthernet 0/0
R2(config-if)# ip address 172.16.0.1 255.255.0.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

2. Verify interface configuration

Command (on both routers):

show ip interface brief

3. Display routing table before static routing

Command:

show ip route

4. Configure static route on Router R1

(To reach 172.16.0.0 network via R2)

Command:

R1(config)# ip route 172.16.0.0 255.255.0.0 10.0.0.2

5. Configure static route on Router R2

(To reach 192.168.1.0 network via R1)

Command:

R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1

6. Verify static routes

Command:

show ip route

*(Look for routes marked with **S**)*

7. Configure IP address on PC1

Command (PC Command Prompt):

ip 192.168.1.100 255.255.255.0 192.168.1.1

8. Configure IP address on PC2

Command:

ip 172.16.0.100 255.255.0.0 172.16.0.1

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9. Test network connectivity

Command:

ping 192.168.1.100

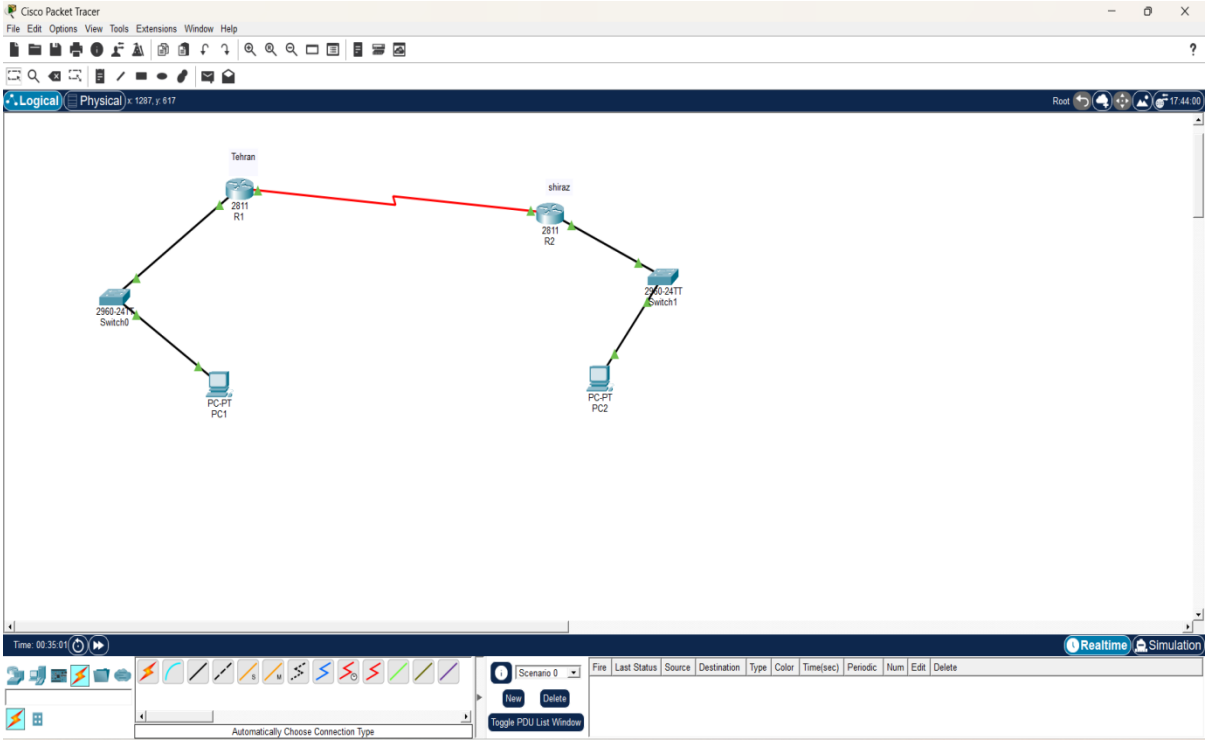
(or)

ping 172.16.0.100

Result

Successful ping confirms that **static routing is correctly configured** and both networks can communicate.

Solution:



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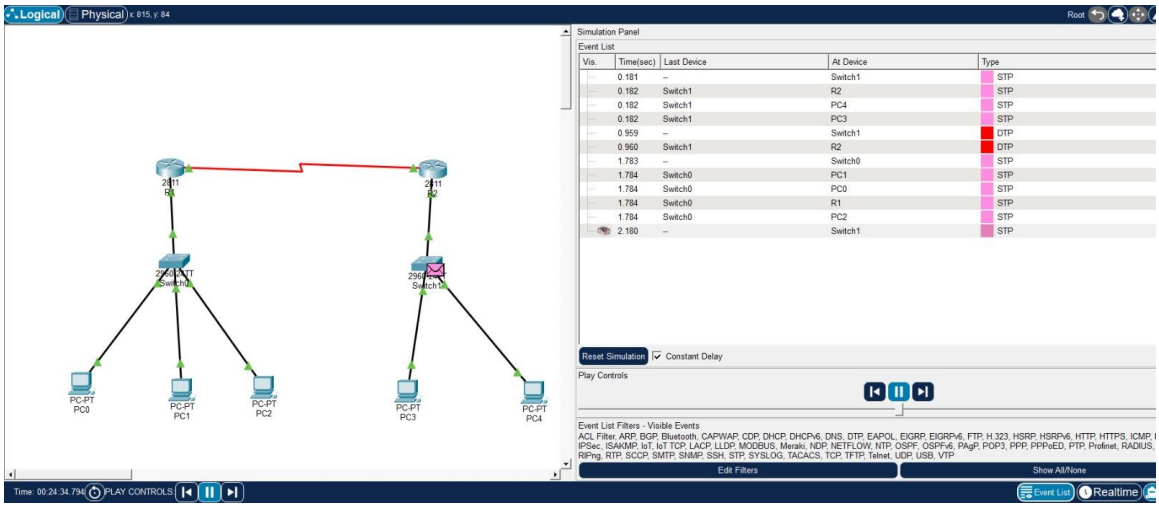
Q.2 Suppose you want to create 3 subnets of a network having network address 198.168.1.0. In another network with network address 220.168.2.0, you want to create 2 subnets. For this scenario configure the end devices and routers accordingly. Connect two different networks (both are subnetted) through static routing and show the message transmission among hosts of the two networks.



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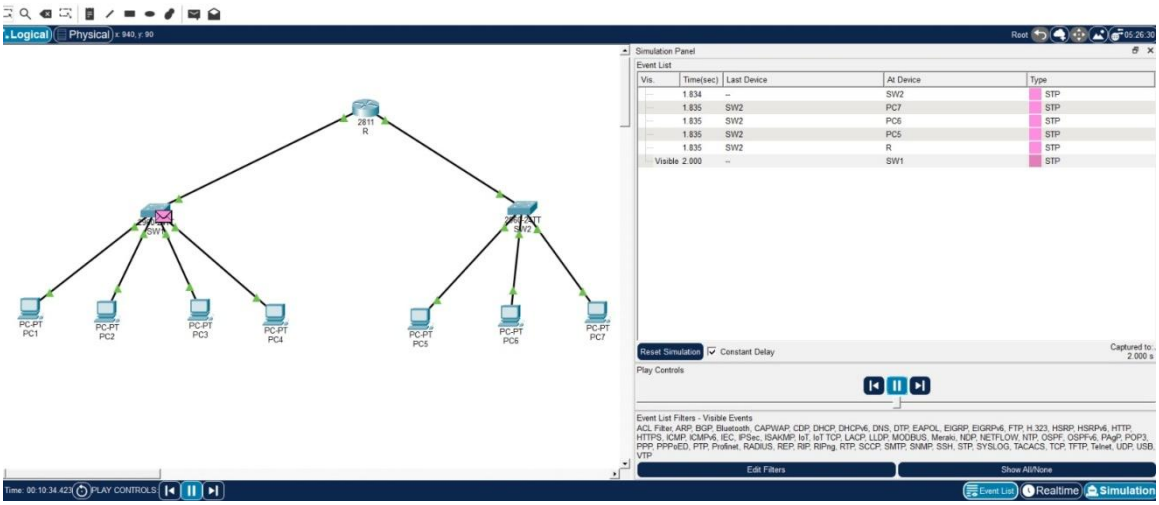
Q.3. Connect two networks having network address 192.168.3.0 and 224.152.1.0 respectively. Network 1 having 4 and network two having 3 hosts. Connect two different networks through a router. For this scenario configure the end devices and routers accordingly and show the message transmission among hosts of the two networks.

Solution: